

Synthetic options, fully collateralised, settled on a sampled median

Every position is collateralised in full the moment it exists, so there is no margin, no liquidation and no counterparty that can fail. One leg of a series is an option and the other is what the deposit becomes once that option has been split off it. A series settles on the median of spaced samples drawn from an on-chain venue, and a matched pair recovers its deposit at par at any time, including when settlement never happens.

■ O , the option leg ■ N , the note leg, fully secured $O + N = 1$

Abstract

Longbow splits one deposit of collateral at a strike into two claims whose payoffs sum to that deposit at every price. Only one of them is an option. The other is the residual that option leaves behind, and both are collateralised in full at the instant they are created, so no participant can owe more than they hold, there is no margin anywhere, and no account can be liquidated. The option is synthetic in the literal sense: no seller writes it, it is produced out of a deposit, and that deposit is the whole of the obligation standing behind it. Two constructions cover the market: a call secured in the asset, and a put secured in cash. Which of the two secures a series is not a design choice. A payoff can be collateralised only in a numeraire that bounds it, and these two are bounded in opposite ones.

Series expire and settle on a number. That number is the median of spaced samples drawn from an on-chain venue across a window that closes at expiry, with samples below a depth floor discarded rather than averaged in, and a minimum count below which the series does not settle at all. A median is chosen over an average because an average moves with one print in proportion to its size, while a median moves only for someone who controls more than half the samples, which means holding a false price across most of the window while anyone who can see it trades against them.

Once the number is fixed each leg redeems for its own share of the deposit, in collateral, and the two shares sum to one deposit exactly at any settlement value whatever. So the escrow holds what it owes however the price turns out, and a price that is wrong moves value between the two legs of one series rather than opening a hole in the collateral. A matched pair needs none of this: it recovers its deposit at par at any time, before or after expiry, whether or not settlement ever happens. That is the one exit that depends on no oracle, no venue and no operator.

Minting and merging a pair are free, unrestricted and symmetric, which pins the two market prices to $0 + N = 1$ in units of collateral to within the cost of gas, so one market prices both legs rather than two markets pricing them inconsistently. The canonical market quotes the option leg against its own collateral, where its price is provably bounded in the unit interval for the whole life of the series, and a bounded domain is the best case for concentrated liquidity. What the design does not remove is stated at full strength in chapter 10: a well-funded party can move a sampled median, an arbiter can set a value on a series whose oracle never finalises, full collateralisation costs the netting that margin buys, and a passive quote near expiry is adversely selected by anyone who can see the strike being crossed.

Contents

1	Notation
2	The primitive
3	Instruments
4	The four operations
5	The life of a series
6	Settlement
7	The market
8	Why nothing can go insolvent
9	Parameters
10	What this does not do
11	What is deferred
A	Worked example

01 Notation

Every symbol used anywhere in this document is defined here, before first use. Nothing below introduces a variable that is not in this table.

A **series** is the unit everything else is said about: one underlying, one strike, one expiry, one settlement configuration and one of the two collateral classes. Creating a series fixes all of them and they are immutable afterwards. A **deposit** is one unit of whatever secures the series, which is one unit of the asset on a call series and S in cash on a put series, and every quantity in this document is counted in deposits so that the two constructions can be said once instead of twice.

Two modelling assumptions are stated here rather than left implicit. The interest rate is taken as zero, so a series carries no cost of carry on either collateral class and moneyness is read against spot rather than against a forward. It is also why both payoffs can be written as they are below, with no discount factor between the strike and what a leg redeems for. The second is a volatility, which appears in the two figures that draw a premium and nowhere else. The protocol computes no option value at any point, so nothing in the design turns on what it is.

$$r = 0, \quad \sigma = 0.70$$

$r = 0$ assumed throughout: no carry on either collateral

$\sigma = 0.70$ in the figures that draw a premium, and in nothing the design reads

SYMBOL	CODE	MEANING
x	x	Price of the underlying. Read at creation to place the series on the grid, and sampled again across the settlement window.
x_T	x_T	Settlement value of a series: the median of the samples taken across the window that closes at expiry. It is the one number both legs redeem against.
r	r	The interest rate, taken as zero throughout. Stated so that the one assumption every figure rests on is visible.
S	S	Strike of a series, fixed at creation.
$\kappa(a, b)$	kappa(a, b)	The payoff kernel, min(1, a/b) .
a, b	a, b	The two arguments of κ . The denominator b is the numeraire the series is secured in: S in cash on a put series, one unit of the asset priced at x on a call series.
O, N	O, N	The two legs of a series: the option leg, which is the option, and the note leg, which is the residual it leaves behind. Each names a share of one deposit, which is why the pair sums to one. The contracts name the same two legs P and N the other way round, so the mapping is stated once here and this document keeps O and N throughout.
O_{out}, N_{out}	O_out, N_out	How many of each leg are outstanding, in deposits. Kept apart from O and N on purpose: a count is what chapter 4.1 conserves, a share is what chapter 4.2 pays, and one symbol cannot carry both.
σ	sigma	Volatility. It appears in the two figures that draw a premium and nowhere in the design, since the protocol computes no option value at any point.
B	B	Sample cap. The most samples a series can hold, which bounds the one-time sort and with it the count anyone can raise an attacker's threshold to.
T	T	Expiry of a series. Minting stops here and the sampling window closes here.
W	W	Length of the sampling window, which ends at T . A sample may be recorded in [T - W, T] and at no other time.
u	u	Minimum spacing between two recorded samples. It is what forces the window to be covered rather than filled in one block.
c	c	Minimum samples a series needs before it can settle at all. Below it the series does not settle and the pair exit is what remains.
L	L	Depth floor. A sample drawn from a venue thinner than this is discarded rather than recorded.
δ	delta	Trading cutoff. The canonical market stops quoting at T - delta , before the option leg becomes a step function that a passive quote cannot survive. It is a venue rule and it does not touch sampling, which reads a different venue.
t	t	Elapsed time, measured from creation.
n	n	Deposits minted over the life of a series, cumulative.
m	m	Pairs merged, cumulative.
q	q	Quantity of a single operation, in deposits, since one deposit is what mints one of each leg.
α	alpha	Collateral the escrow holds, in deposits. The asset on a call series, cash on a put series.
M	M	Seed minimum. The smallest deposit that may create a series, locked as liquidity until the cutoff.
g	g	Ratio of the strike grid. Admissible strikes are the reference mark times integer powers of g .
k	k	Step on the strike ladder, any integer. The one symbol here that counts rungs rather than quantities.

Cash means the dollar stablecoin a series is quoted in, and the asset means the ERC-20 the series is written on. Both are ordinary tokens with no behaviour of their own, which is a constraint on what may be listed rather than an assumption about what exists: chapter 10 says what that rules out.

02 The primitive

One function generates both constructions in this document. Which of the two a series is, and what secures it, follows from that function's arguments rather than from a decision.

2.1 One kernel

Deposit collateral, receive two claims that always sum back to it.

$$\kappa(a, b) = \min(1, \frac{a}{b}), \quad N = \kappa(a, b), \quad O = 1 - N$$

$$\text{kappa}(a, b) = \min(1, a/b) \quad N = \text{kappa}(a, b) \quad O = 1 - N$$

The note leg is always κ of two arguments and the option leg is always the residual, which is what makes the sum exact by construction rather than by arithmetic that could round wrong. Everything else in this document follows from choosing what a and b carry, and that choice is not free.

2.2 Why the payoff chooses the collateral

An uncapped call pays $\max(0, x - S)$, which is unbounded above, and a fixed quantity of cash is not. Uncapped upside cannot be collateralised in cash at all. The mirror constraint is exactly as strict. A put pays $\max(0, S - x)$, which S in cash covers at every price, but measured in units of the asset the same payoff is $\max(0, S - x) / x$, and that grows without bound as the price approaches zero, so a put cannot be collateralised in the asset either.

	in cash	$\max(0, S - x) \leq S$	$\max(0, x - S) \rightarrow \infty$
	in the asset	$\frac{\max(0, S - x)}{x} \rightarrow \infty$	$\frac{\max(0, x - S)}{x} \leq 1$

	the put leg	the call leg
in cash	$\max(0, S - x) \leq S$	$\max(0, x - S)$ unbounded
in the asset	$\max(0, S - x)/x$ unbounded	$\max(0, x - S)/x \leq 1$

So the collateral is not chosen: a payoff can be secured only in a numeraire that bounds it, and each of these two is bounded in exactly one. That fixes the collateral of every series in the system, and it is the same fact that makes liquidation unnecessary, since every position is bounded by the collateral behind it.

2.3 What mint and merge pin

A deposit may be split into a pair, and a pair may be burned back into a deposit, by anyone, for gas. Neither operation has a fee, a permission or a counterparty. That symmetry is the load-bearing property of the whole design, because it puts a two-sided arbitrage on the pair of market prices: if the pair trades above a deposit, mint and sell it, and if it trades below, buy it and merge.

$$O + N = 1$$

$$O + N = 1 \quad \text{in deposits, at every price and after every operation}$$

Two consequences follow and both are used later. One market prices both legs, since the price of the note leg is one minus the price of the option leg and needs no venue of its own. And the identity is a property of the construction rather than of the market: a pair is one deposit at every price and in every phase, so it is not something the arbitrage establishes and could stop establishing. What the arbitrage does is hold the two quoted prices on it, and the merge side of that trade is open from the first block to the last.

03 Instruments

Only one leg of a series is an option. The other is what the deposit becomes once that option has been split off it, and there is no protocol-specific vocabulary to learn for either.

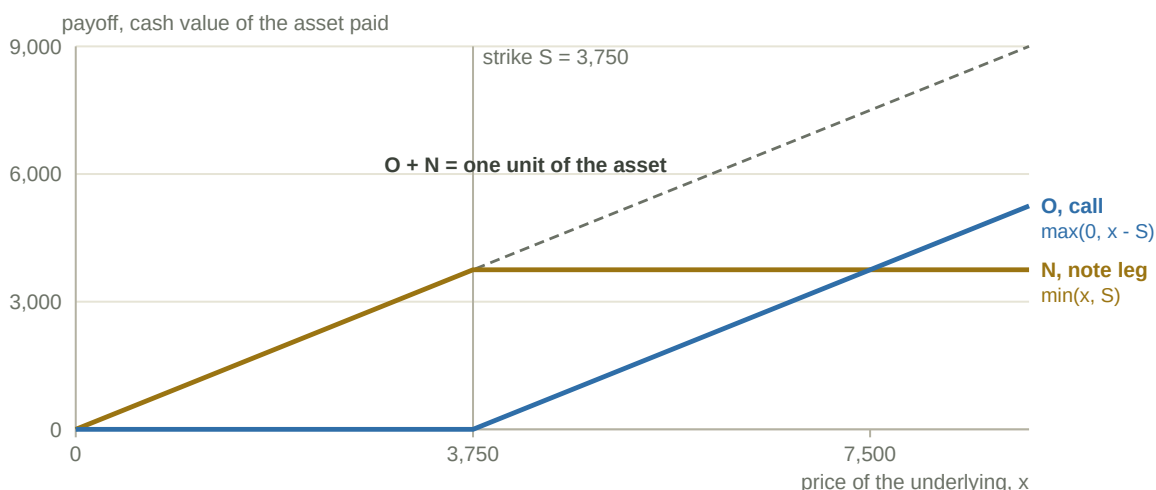
SERIES	COLLATERAL	OPTION LEG IS	NOTE LEG PAYS LIKE
Call at S	one unit of the asset	CALL	COVERED CALL
Put at S	S in cash	PUT	CASH-SECURED PUT

The right-hand column says *pays like* and not *is*, and the distinction is the whole of what makes these claims synthetic. A covered call is a position somebody assembles by holding spot and selling a call to a buyer. The note leg is what is left of a deposit after an option has been split off it, with no sale, no buyer and no obligation anywhere. The payoffs coincide. The arrangements behind them do not.

3.1 The call series

$$N = \kappa(S, x), \quad O = 1 - \kappa(S, x)$$

$N = \kappa(S, x)$ $O = 1 - \kappa(S, x)$
in cash, per asset unit deposited: N pays $\min(x, S)$ O pays $\max(0, x - S)$



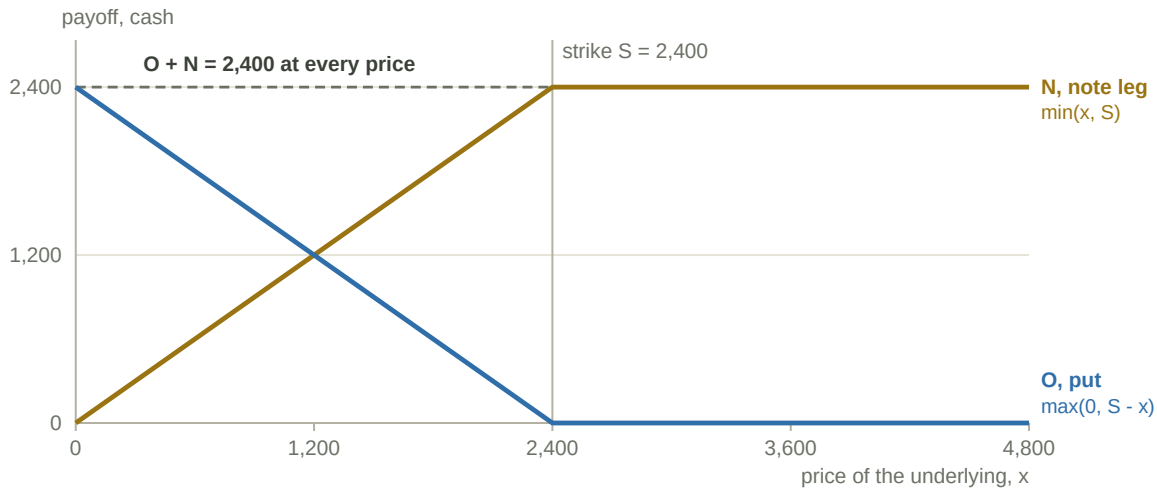
The call series. One unit of the asset deposited, two payoffs that sum to that unit at every price. Both legs are paid in the asset, so the axis is the cash value of the asset each one receives. The pair sums to the diagonal, because what it adds back to is the asset, whose cash value moves with the price.

The option leg is a call, uncapped. The note leg is not a second option: it is the deposit with that call removed from it, and what is left pays $\min(x, S)$, which is what a covered call pays. Nobody wrote the call, so there is no covered-call seller anywhere in the arrangement and no obligation that could fail. The asset unit is in the escrow before either leg exists.

3.2 The put series

$$N = \kappa(x, S), \quad O = 1 - \kappa(x, S)$$

$N = \kappa(x, S)$ $O = 1 - \kappa(x, S)$
 in cash, per S deposited: N pays $\min(x, S)$ O pays $\max(0, S - x)$



The put series. One deposit of S , two payoffs that sum to S at every price. Both legs are paid in cash here, since cash is what secures the series. The note leg tracks the underlying down and flattens at the strike; the option leg is what is left.

These are the call series' arguments the other way round, and that swap is the whole difference. The option leg is a put, and the note leg is the deposit with that put removed from it, which pays what a cash-secured put pays. The same absence of a writer holds: the cash is in the escrow before either leg exists, so nothing about the arrangement can be short of it later.

3.3 What a buyer gets

Both note legs pay $\min(x, S)$, and that single expression is the honest description of what the residual is. It flattens at the strike, which is what the note leg is sold at a discount for, and below the strike the note leg **carries the full downside below the strike**, exactly as the underlying does. Bought at a discount to its cap it is a fixed-target yield instrument, and the target is reached in every state except the one where the underlying falls, which is not a small exception and is not described here as one.

The option leg is the mirror. It costs a premium that is entirely at risk, it expires worthless whenever the strike is not crossed, and it is uncapped on the side it is written on. A holder who wants exposure without the premium at risk is describing the note leg, and a holder who wants convexity is describing the option leg. There is no third thing in the system.

A COVERED CALL PAYOFF IS MINTING AND SELLING ONE LEG

Deposit an asset unit, receive both legs, sell the option leg and keep the note leg. That is the covered call payoff produced out of two operations, and it is the flow the design expects most supply to arrive through. Nothing is written along the way. The deposit is the whole of the obligation and it is in the escrow before either leg exists, so nothing about it can be under-collateralised later, and there is no margin call to answer because there is no margin.

04 The four operations

Mint, merge, settle and redeem. Everything the escrow can be asked to do is one of these, and only one of them reads anything from outside.

OPERATION	WHEN	IN	OUT
Mint	$t < T$	q deposits of collateral	q of each leg
Merge	any t	q of each leg	q deposits of collateral
Settle	$t \geq T$, once	nothing	the settlement value, fixed for the series
Redeem	after settlement	q legs of either kind	that leg's share of q deposits

Settle moves no tokens and takes no argument. It is permissionless, it may be called once, and all it does is copy the number chapter 6 produces into the escrow so that redemption has something to divide against. Merge is the only operation with no time bound at all, which is chapter 5.3.

4.1 One conservation statement, not two

The escrow holds one thing: α , the collateral it was given. There is no second pot, because no operation pays anything in except a mint, and a mint pays in exactly one deposit per pair it creates.

α	$= n - m$
O_{out}	$= n - m$
N_{out}	$= n - m$
alpha = n - m	$O_{\text{out}} = n - m$ $N_{\text{out}} = n - m$

Before settlement those three lines are the same number, so every deposit in the escrow stands behind exactly one live pair. After settlement the two legs are redeemed separately and the counts come apart, and what holds the statement together then is the kernel itself: whatever the settlement value turns out to be, the two shares sum to one.

$\alpha = N_{\text{out}} \cdot \kappa + O_{\text{out}} \cdot (1 - \kappa)$
alpha = N_out * kappa + O_out * (1 - kappa) before and after every operation, for any kappa

The collateral in the escrow equals what the escrow owes, at every instant, under every interleaving of mints, merges, settlement and redemptions, and for any settlement value whatever. That last clause is the one worth pausing on. The identity does not depend on the price being right. A settlement value that is wrong moves collateral between the two legs of that one series and cannot make the escrow short, because κ and $1 - \kappa$ sum to one however κ was computed. This is the non-liquidatable claim written as arithmetic rather than as a policy, and chapter 8 draws the consequences out.

4.2 What each leg redeems

A single number each, in the collateral the series is secured in. Not a share of a pot that other holders' behaviour filled, and not a pair of holdings in two different denominations: one number, fixed for the whole series the moment settlement happens.

note leg : $\kappa(a, b)$ deposits
option leg : $1 - \kappa(a, b)$ deposits

per note leg: kappa evaluated at x_T , in deposits
per option leg: $1 - \text{kappa evaluated at } x_T$, in deposits

Every note leg of a series is worth exactly what every other one is, and the same holds for the option legs, because nothing about a redemption depends on when it happens or on who went first. Both shares are rounded down independently, so a pair redeemed leg by leg recovers marginally less than the deposit it came from and the rounding always favours the escrow rather than the last holder out. A holder who wants the rounding not to happen at all merges instead, which returns the deposit whole.

05 The life of a series

Two phases and two markers inside the first one. Only one boundary in this chapter closes an extraction, and the chapter is honest about which.

5.1 The phases

	active	sampling	no quote	settled
mint	yes	yes	yes	
merge	yes	yes	yes	yes
trade	yes	yes		
sample		yes	yes	
settle				yes
redeem				yes
		$T - W$	$T - \text{delta}$	T

merge is the one row open in every stretch, which is the exit that consults nothing

The life of a series. Minting runs to expiry and the canonical market stops a little before it. The sampling window closes at expiry, settlement follows once the window has closed, and merging is available from the first block to the last.

The **active phase** runs from creation to expiry: mint, merge and trade, with the market stopping at the cutoff $T - \text{delta}$ a little before expiry itself and the sampling window opening at $T - W$. At expiry minting stops and the sampling window closes. The series is **settled** once the samples have been reduced to a median and that number copied into the escrow, after which redemption is open and stays open. Merging is available throughout, in every phase, and is the subject of chapter 5.3.

5.2 Why minting stops at expiry

Minting is value neutral at every instant of a series' life. A pair is worth one deposit whatever the price is doing, so a minter pays a deposit and receives claims worth a deposit, and no incumbent holder is diluted by it. That is worth stating plainly because it means the boundary at expiry is not closing an extraction, and a document that dressed it up as one would be inventing a defence the design does not need.

What the boundary does is keep the instrument honest about what it is. After expiry a series has no optionality left to sell, and its settlement value is either already fixed or determined by samples that have already been taken. Minting into it

would create fresh claims on an outcome that is decided, which is not theft but is not an option either. The gate is there so that every pair in existence was created while the thing it is a claim on was still uncertain.

The same reasoning is why minting stays open through the sampling window. Samples begin at $T - W$ while the series is still live, and a minter during that stretch is buying a pair whose value is still one deposit, so nothing about the window needs to be defended against.

5.3 Why merging never stops

A matched pair is one deposit by construction, at every price, in every phase, before settlement and after it. So returning a pair and taking a deposit can never be paid for out of anybody else's claim, and the escrow applies no time bound to it at all. This is the operation that makes the system's worst cases survivable rather than merely unlikely.

THE EXIT THAT CALLS NOTHING

A holder of both legs recovers the deposit at par whether or not the oracle ever finalises, whether or not the venue the samples come from still exists, and whether or not any operator anywhere does anything. Merge reads no price, consults no configuration and asks no permission. It is the reason a series that fails to settle is a series whose holders are stuck with each other rather than a series whose collateral is stuck in a contract.

Fewer boundaries is the point. A rule that exists because an operation would otherwise pay for itself out of somebody else's claim is a rule that can be written wrongly, and the way not to write it wrongly is for the operation not to have that property in the first place. Merge does not have it, at any price, in any phase, so it needs no rule and does not have one.

06 Settlement

The protocol has to learn one number per series. Everything in this chapter is about making that number expensive to move, cheap for anyone to check, and impossible to choose by timing.

6.1 A median of spaced samples

Across the window $[T - W, T]$ anyone may record a sample of the underlying's price, drawn from an on-chain venue fixed for the series at creation. A sample is refused if it comes less than u after the last one, and refused if the venue backing it is thinner than the depth floor L . When the window closes the samples are sorted once and the median is stored. On an even count the two central samples are averaged, and both of them are recorded observations, so the stored number stays inside the range the window actually saw. That number is the settlement value, and copying it into the escrow is the whole of what settle does.

$$x_T = \text{median}(\text{samples in } [T - W, T])$$

x_T = median of the samples recorded in $[T - W, T]$
each sample at least u after the last, each from a venue at least L deep,
and at least c of them or the series does not settle

Three properties fall out of that shape and each of them is deliberate. Sampling is permissionless by default, and in that form no operator is trusted to perform it and holders on both sides have an interest in keeping the window populated. A series may instead name, at configuration, the only addresses that may record into it, and that set is fixed for its life. Which of the two forms a series takes is decided at creation and visible before anybody mints into it, and what the restricted form buys and what it costs is chapter 10. The window closes at expiry rather than opening there, so settlement is decided by the life of the series rather than by a stretch of time after it in which nothing is at stake. And nobody chooses the moment: the

median of a fixed window cannot be picked by timing a call, which is the failure that a single reading taken at expiry has and this does not.

6.2 Why a median and not an average

An average moves with every print in proportion to that print's size and how long it stands, so one block of borrowed liquidity moves it, and the amount it moves it by is a smooth function of how much was borrowed. There is no threshold to clear. A median has one. Capturing a minority of the samples moves it only within the range the honest samples occupy, since what it selects is still one of them. Moving it outside that range takes more than half of the recorded samples, and the spacing rule turns that into an amount of time rather than an amount of capital.

$$t \geq u \cdot \lfloor c / 2 \rfloor$$

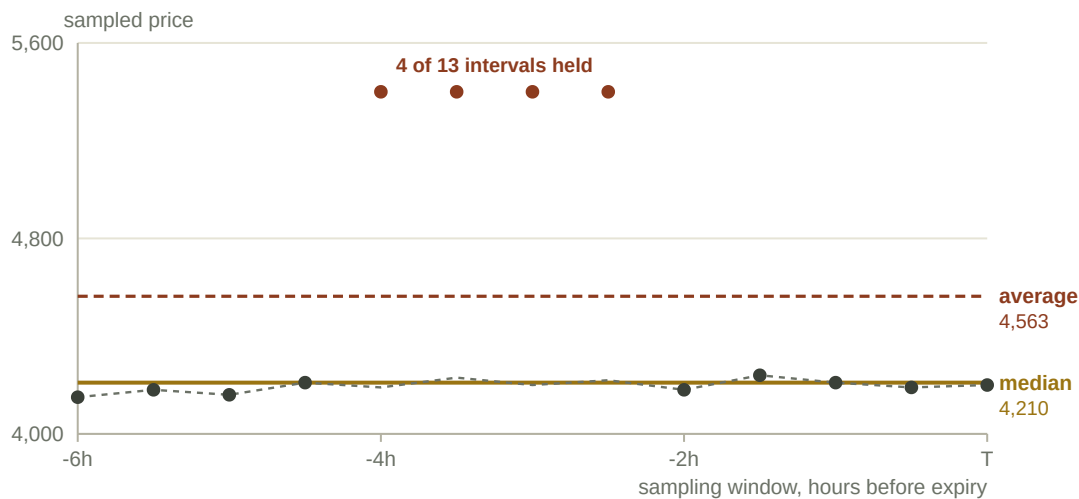
a strict majority of c samples is $\text{floor}(c/2) + 1$ of them,
which spans $\text{floor}(c/2)$ intervals at the enforced spacing

with $c = 12$ and $u = 15$ minutes:

7 of the 12 samples must be yours, spanning 6 intervals,
so the false price has to stand for at least 90 minutes

That time cost has a condition attached, and it is worth stating rather than leaving to be found. The spacing is enforced per series and not per submitter, which is the right way round: a per-submitter rule would let one party record from many addresses at the same instant. What it means is that every interval is a race and the party who records first takes it. In a window that others contest, a false price has to stand rather than merely appear, because a sample taken at any moment in between lands at the honest price. A window nobody else touches is a different thing. The attacker takes every interval unopposed and needs the venue dislocated only at the instants they choose to record, so the spacing then bounds how long the attack takes to run rather than how long the false price has to be held. The defence is other people sampling, which is the same permissionless property that makes it nobody's duty. On a series that restricts recording to a named set, that defence is worth whatever that set is worth and no more, since there is nobody else to contest an interval.

The figure below puts numbers on the difference. A thirteen-sample window in which four intervals are held at 5,400 against an underlying trading near 4,200 settles at 4,210 on the median, ten above where it should, and at 4,563 on the average, three hundred and sixty three above. The same attack buys thirty six times as much against the average as against the median, and what the four bought against the median was a step from one honest sample to another rather than any part of the move to 5,400. Settling anywhere near the price they printed would have taken seven of the thirteen. Ninety minutes of holding a price away from where the rest of the market thinks it is, against everyone who can see it and trade the difference, is a cost that scales with the deviation and with the depth of the venue. It is not a guarantee and chapter 10 does not describe it as one. It is the difference between an attack that costs a flash loan and an attack that has to be financed and defended in the open, and every additional sample anyone else records raises the count the attacker has to beat.



What the median discards. Thirteen spaced samples with the underlying drifting, four of the intervals held well away from it. The average follows the excursion in proportion to its size. The median barely moves, because a minority of captured samples can only select a different honest sample and never the price they printed.

6.3 The depth floor and the count

Two refusals do the rest of the work, and both are refusals rather than adjustments. A sample drawn from a venue thinner than L is discarded and not recorded at all, which closes the cheapest version of the attack: drain the venue first, then print into the vacuum at no cost. And a window that closes with fewer than c samples does not settle the series. The design would rather leave a series unsettled than settle it on evidence it does not have, which is a live cost to holders and is written down as one in chapter 10.

A third bound is structural rather than a refusal. A series holds at most B samples, which is what keeps the one-time sort cheap, and it is also the ceiling on the last claim of chapter 6.2: extra samples raise the count an attacker has to beat only up to B . At the parameters of chapter 9 the window cannot produce that many, so the cap is slack there, which is a reason to publish it rather than a reason to leave it in the code.

6.4 When settlement does not happen

A series whose window never filled, or whose venue stopped being readable, has no settlement value and therefore no redemption. Two things stand against that, and they are of very different strengths.

The first is the pair exit of chapter 5.3, which is unconditional. Anyone holding both legs takes the deposit back at par, and no oracle, venue or operator is consulted on the way. The second is an arbiter who may set a value on a series that has failed to settle, after a grace period has passed. That power is real, and it is bounded in one way worth relying on and in one way weaker than it first reads. It closes once the oracle has finalised, because settlement is then open to anyone and this path is shut against a value that exists. It does not close the moment a window holds enough samples to finalise on: between the grace period elapsing and somebody calling finalise, a series that would have settled honestly is still reachable. So the arbiter can never override a settlement that exists, and the way to make that bound bind is to make the settlement exist.

THE ARBITER IS A TRUSTED PARTY AND IS NAMED AS ONE

On a series that has not settled, the arbiter chooses the number both legs are paid against, and nothing in the arithmetic constrains what that number is. It cannot make the escrow short, because the two shares still sum to one deposit, but it decides how a series' collateral is divided between its holders. A holder who is unwilling to rely on that has one answer available to them and it is the pair exit.

07 The market

The escrow needs no venue. The venue exists because a holder who wants one leg rather than both needs someone to take the other, and it is built where a bounded price is worth the most. This chapter describes a design and not a deployment: the escrow is written, the venue is not.

7.1 One pool prices both legs

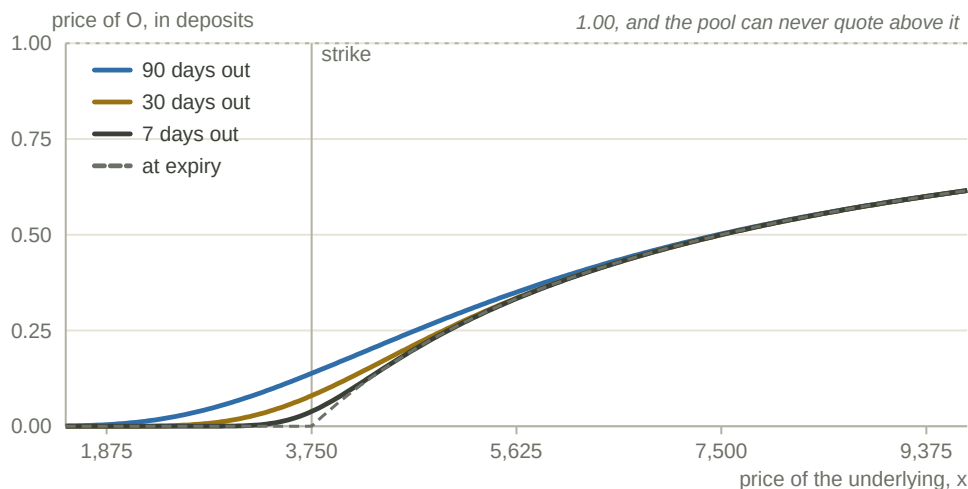
Because minting and merging pin $0 + N = 1$, a quote on either leg is a quote on both, and a single pool is enough. The canonical market for a series is the pool that quotes the option leg against the collateral of its own series. A note leg is priced by subtraction, and the escrow places no restriction on where either leg changes hands. What stands in the way of a second venue is not the escrow but the form the legs take: identifiers inside one contract rather than tokens an exchange already speaks, which is the wrapper chapter 11 names as required.

7.2 The domain is bounded, which is why a seed can work

An option is never worth more than the thing it is written on and never worth less than nothing. Quoted in units of its own collateral, the option leg therefore lives in the unit interval for the entire life of the series, and this is a fact about the payoff rather than a range someone chose.

$$0 \leq \text{price of } O \leq 1$$

$0 \leq \text{price of } O \leq 1$ in deposits, for all t



The pool's whole domain. The option leg quoted in deposits, against the price of the underlying, at three distances from expiry and at expiry itself. Every reachable price lies in the band, so seeded liquidity is never stranded where the market cannot go.

A market maker on an unbounded pair has to decide which prices are worth quoting and accept that capital placed outside that guess does nothing. Here there is no guess: the feasible domain is the whole domain, so a seed deployed across it is fully in play from the first swap. That is what makes a market that is created at the same instant as the instrument a reasonable thing to expect to work.

7.3 Where the system reads a price

Two places, and they are different venues doing different jobs. At creation the underlying's own market is read once to place the series on a grid, because unconstrained parameters would scatter liquidity across strikes and expiries that nobody can find. Expiries sit on a fixed schedule and strikes at the reference mark times integer powers of g .

$$S \in \{x \cdot g^k\}, \quad g = 1.25$$

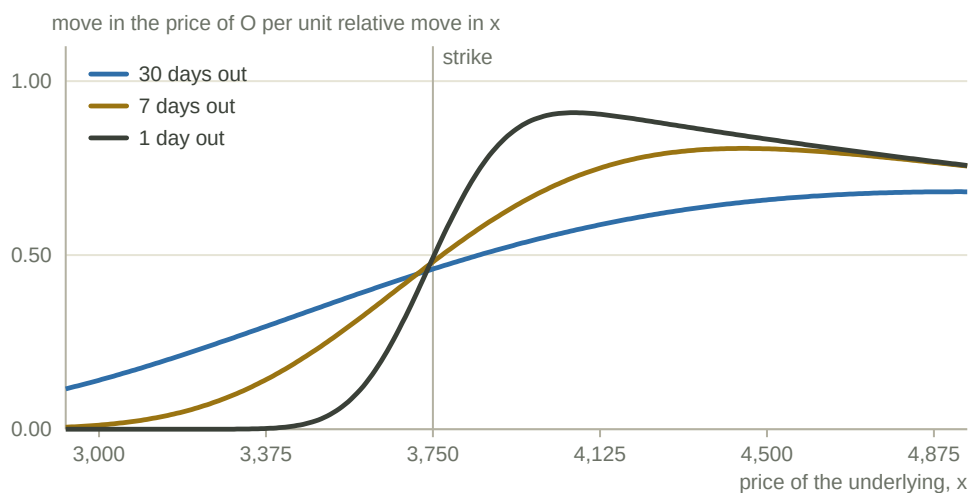
admissible strikes: $S = x \cdot g^k$ for integer k , with $g = 1.25$
 read once, at creation, when the series holds nothing

That reading is the harmless one. It happens when a series holds nothing and nobody is exposed to it, and the worst a manipulated mark can do is place a series on an odd strike that other people then decline to trade. The second reading is settlement, which is chapter 6, and it is the one that matters: it decides how collateral is divided, so it is sampled rather than read, spaced rather than instantaneous, and floored on depth rather than taken on trust.

The two venues are deliberately not the same one. Settlement samples the market in the underlying itself, behind a source interface that hides which venue that is, so a series settles on the asset's own market rather than on the market in its own claims. A pool quoting the option leg is thin by construction, since it exists only for one strike and one expiry, and settling a series on the price of its own claims would put the cheapest venue in the system in charge of the largest number in it.

7.4 The cutoff, and what it admits

Close to expiry an at-the-money option leg stops behaving like a curve and starts behaving like a step: worth nearly nothing on one side of the strike and nearly its full intrinsic on the other, with the crossing happening in whatever time it takes the underlying to move a little. A passive quote cannot survive that. Anyone who can see the strike being crossed trades against the curve and is right every time.



Why quoting stops before expiry. The option leg's sensitivity to the underlying, at three distances from expiry. What is a gentle slope a month out is very nearly a step on the last day, and a passive curve has to hold still against it, which is what the cutoff refuses rather than prices.

The design's answer is to stop rather than to charge for it. At $T - \text{delta}$ the canonical market halts, and the last stretch before expiry has no quote at all. This is the honest version of a hard problem: there is no passive curve that is safe against informed flow at the money on the last day, so the venue does not pretend to offer one. What it costs is a blackout in which a holder can wait but cannot sell, and chapter 10 counts that as a cost rather than a feature. The cutoff does not touch sampling, which reads the underlying's market and is unaffected by whether the pool in the claims is quoting.

7.5 The seed, and what it is locked for

Creating a series requires committing at least M of collateral to the canonical market. Part of it is split, and the option legs it produces are placed against the collateral that was not, since the pool of chapter 7.1 quotes one against the other. The note legs stay with the creator, which is the covered call flow of chapter 3 with the sale made into their own position. The lock runs to the cutoff, at which the market is frozen anyway, so nothing is bought by holding the position past it.

What the lock buys is that a series which exists has a market for as long as it is tradable, which is a claim a taker can rely on rather than a hope. The creator is paid for it in the swap fees the position accrues, and M doubles as a defence against

series spam, since creating one costs real capital for a real duration.

08 Why nothing can go insolvent

The claim is not that the system is well managed, and it is not that nothing can go wrong. It is that a specific list of failures is unreachable, and the reason is one line of arithmetic that every operation preserves.

Mint raises n by q , which raises α and both leg counts by q each. Merge raises m by q , which lowers all three by q each. Settle moves nothing. Redeem burns a leg and pays out exactly that leg's share, lowering α by what it paid. No operation moves one side without the other, so the equality is not maintained by a check that could be forgotten: it is the only thing the operations can do.

$$\alpha - (N_{\text{out}} \cdot \kappa + O_{\text{out}} \cdot (1 - \kappa)) = 0$$

alpha - what the escrow owes = 0 before and after mint, merge, settle and redeem

Four things follow, and each of them is a class of failure that other designs have to defend against and this one cannot reach.

No redemption can fail. Every pair has its deposit sitting in the escrow and the two shares sum to that deposit, so the question of whether some writer is good for it has no content. There is no writer. There is nothing to liquidate because nothing can become under-collateralised, and no keeper needs to be paid to watch for it.

A wrong price cannot make the escrow short. This is the one that has to be stated precisely, because a price is read and it can be wrong. A settlement value that is too high or too low changes how a series' collateral is divided between its own two legs and changes nothing else, since κ and $1 - \kappa$ sum to one however κ was arrived at. Manipulation is a transfer between holders of one series. It is a real cost to the side that loses it, chapter 10 says so, and it is not a hole in the collateral.

A series cannot reach another series. Each one is accounted against its own outstanding deposits, so what a series can pay out is bounded by what was minted into it, and that holds even where two series are written on the same token. A defect in one is bounded by that series.

The venue cannot take the escrow. Mint, merge and redeem never call a market. A pool that is misconfigured, drained or paused is a market that is gone, which is a real cost to holders who wanted to trade and is not a loss of collateral to anyone. A venue that stops being readable delays settlement rather than moving anything, and the pair exit survives both.

WHAT IS LEFT IS THE ORDINARY TOKEN BOUNDARY

The escrow moves value on the mint and redeem paths, which is the classical place for a reentrant token to be given a second look at inconsistent state. Checks are ordered before effects, effects before interactions, and a guard covers every path that moves value. A mint credits the balance the escrow actually received rather than the amount it was asked for, so a token that takes a cut in transit cannot cause claims to be minted against value that never arrived. This is a small, known and reviewable surface, which is the point of having deleted the others, and chapter 10 records that it has not yet been reviewed.

09 Parameters

Everything a series is, fixed once at creation. What is not fixed at creation is privileged, and it is listed here rather than left for a reader to discover.

PARAMETER	SYMBOL	SET BY	PROPOSED
Strike	S	the grid, from the reference mark	$x * g^k$
Grid ratio	g	venue policy, fixed	1.25
Expiry	T	the schedule	weekly and monthly
Sampling window	W	per series, at creation	6 hours, closing at expiry
Minimum spacing	u	per series, at creation	15 minutes
Minimum samples	c	per series, at creation	12
Depth floor	L	per series, at creation	per underlying, measured
Sampling access		per series, at creation	open, or a named set
Trading cutoff	δ	venue policy, fixed	hours, calibrated
Seed minimum	M	venue policy, fixed	per underlying

The settlement four are one trade made four ways. A longer window, wider spacing and a higher count all raise what moving the median costs and all raise the chance that a quiet series fails to reach c and does not settle. A higher depth floor does the same from the other side: it refuses worse evidence and it refuses more of it. Every one of them wants measurement against real flow before launch rather than a number chosen here and defended later.

WHAT IS PRIVILEGED, STATED IN ONE PLACE

These are role slots in the contracts and they are separately assignable. In the pilot they are not separate: one deployer key holds all of them, and it is also the key that registers the series and seeds them. What that key cannot do is change a series. Strike, maturity, collateral, oracle, window, spacing, count, depth floor, venue and the set of addresses that may sample are fixed when the series is created, and no function edits any of them afterwards. The powers that stay live are not parameters.

- An **admin** sets which oracles may be bound to a series, who may list series, and the gate that can halt minting. None of the three reaches a series that already exists, since the oracle is part of its identity.
- A **lister** registers series. Series are not created permissionlessly today, whatever the grid would allow.
- A **configurer** sets a series' sampling window, spacing, count, depth floor and venue, and whether recording is open to anyone or restricted to a set of addresses named at the same moment. It may do so once per series and never again, and no function adds an address to that set or removes one afterwards. The escrow does not require a series to be configured before it can be minted into, so what keeps the configuration out of reach of existing positions is that configuring happens alongside registration rather than a check the escrow makes.
- An **arbiter** may set a settlement value on a series that failed to settle, after a grace period, and loses that power the moment the oracle finalises.
- The **mint gate** can halt new mints. It is never consulted by merge, settle or redeem, so a pause cannot trap collateral.

10 What this does not do

The costs of the design, stated once, in one place, at full strength. Every one of them is the direct price of something claimed earlier in the document.

STATED PLAINLY, AND NOT SOFTENED ANYWHERE ELSE IN THIS DOCUMENT

- **A well-funded party can move a sampled median.** Settlement is a number read off a market, and a market can be pushed. The spacing rule turns the attack into time held rather than capital deployed, and the depth floor refuses the cheapest version of it, but neither makes it impossible. What it costs the side that loses is the difference between the settled value and the true one, on their whole position.
- **The time that costs an attacker depends on somebody else sampling.** Spacing is enforced per series, so every interval goes to whoever records it first. Against contested sampling a false price has to stand across the window. Against a window nobody else touches an attacker takes every interval and needs the price wrong only at the moments they record, which is a different and much cheaper attack.
- **The arbiter can set a value on a series that did not settle.** Nothing in the arithmetic constrains the number they choose. The power closes once the oracle has finalised and not at the moment the samples to finalise on exist, so a series that has cleared its count but that nobody has finalised is still reachable after the grace period. It is unbounded on one that stalled.
- **A series can fail to settle.** If the window closes with fewer than c samples that cleared the depth floor, there is no settlement value and no redemption, and what is left is the pair exit and the arbiter. Sampling is permissionless by default, which means it is nobody's duty: a series that nobody bothers to sample is a series that does not settle. A series that restricts recording to a named set trades that exposure for its mirror rather than escaping it. The duty is then somebody's, the set cannot be added to once the series exists, and if those addresses stop recording the window closes short of c and the series falls to the arbiter, whose number nothing in the arithmetic constrains.
- **The note leg carries the full downside below the strike.** Both constructions pay $\min(x, S)$, so both track the underlying to zero with it, and the upside of each is capped. This is the ordinary risk of the residual side and it is named here rather than described as protection.
- **The contracts are not audited.** Nothing here has been through review. This document is a design, and the venue of chapter 7 and the put construction of chapter 3.2 are described rather than built.
- **There is no market between the cutoff and expiry.** Quoting stops at $T - \text{delta}$ and redemption opens after settlement, so for that stretch a holder may wait but may not sell. Neither leg has a unilateral exit before then either: a note leg can only leave early by buying an option leg to merge against, which needs a counterparty.
- **Full collateralisation buys no netting.** One deposit per option, no spreads offset, no portfolio margin. A trader who is long one series and short another posts collateral on both. This is not a defect to be fixed later: it is what deleting liquidation costs, and cross-series collateral would put the valuation and liquidation machinery straight back.
- **A passive quote near expiry is adversely selected.** The cutoff shortens the window in which that is true and does not remove it. Anyone providing liquidity to a series, the creator included, should expect the fees to be compensation rather than immunity.
- **Liquidity is not a protocol guarantee.** Every strike and expiry is its own market and the grid bounds the fragmentation without ending it. Depth will sit in near-dated, near-the-money series, the wings will be thin, and the seed lock guarantees a market exists and not that it is good.
- **Only plain tokens can be listed.** The escrow holds balances and assumes they stay put: no rebasing, no fee on transfer, no supply multiplier, no distribution to holders. An asset whose balance moves on its own breaks the arithmetic of chapter 4 silently rather than loudly, so what may be listed is constrained at the venue.
- **Series are listed rather than created on demand.** Registration is permissioned today. The grid is what series would be constrained to if it were opened, and it is not the thing standing in the way.

- **The regulatory position is not settled.** Options sit adjacent to regulated derivatives in most places. The protocol is infrastructure and the constraints will land on interfaces and distribution. This document makes no legal claim.

11 What is deferred

Named here so that absence is a decision on the record rather than an omission a reader has to notice.

The put construction. The kernel covers it and chapter 3.2 states it, and the escrow implements one ordering of the kernel rather than both. Nothing in the design has to change to add it, which is why it is described here rather than removed.

The venue. All of chapter 7 is design. The escrow has no dependency on it and works with whatever market exists, but a holder who wants one leg rather than both needs a market to exist, so this is the item between the primitive and anyone actually using it.

A transferable wrapper per leg. The legs are identifiers inside one contract rather than tokens an exchange or a wallet already speaks. Until each side has an ordinary token of its own, a holder of a single leg holds it to settlement unless they can find the other. This is required rather than optional, since without it only people who want both sides can take part.

Vaults that write on a schedule. Minting against a deposit, selling the option leg into the canonical market and rolling at expiry is a strategy built on top of these operations, and it belongs above the protocol rather than inside it.

Series that do not expire. A series funded continuously by decay of the option leg's claim, rather than settled at a date, removes expiry and everything that hangs off it, and replaces it with a funding rate the protocol has to derive and a price it has to read continuously rather than once. That is a different set of trades from the ones this document makes, and it is research rather than roadmap here.

Machine-checked proofs. The conservation statement of chapter 4.1 is the argument worth proving rather than testing, and it should be proved before real funds.

A Worked example

One call series through its whole life, then the same numbers on the mirror construction. Every figure quoted anywhere in this document comes from here.

A series is created on an asset marked at 3,000. The grid ratio of 1.25 puts the nearest strike above the mark at 3,750, and the creator seeds the canonical market with the minimum. Over the active phase 100 deposits are minted, each one asset unit, and 20 pairs are merged back, so 80 pairs stand at expiry. Across the six-hour window closing at expiry, 13 samples are recorded, every one of them clearing the depth floor, and their median is 4,200. That is the settlement value, and the series settles on it.

QUANTITY	CODE	VALUE
Deposits minted	n	100
Pairs merged	m	20
Collateral in escrow	alpha	80 asset units
Option legs outstanding	O_out	80
Note legs outstanding	N_out	80
Settlement value	x_T	4,200
Per note leg	kappa(S, x_T)	0.892857 asset units
Per option leg	1 - kappa	0.107143 asset units

At 4,200 a note leg's 0.892857 of an asset unit is worth 3,750, which is $\min(x_T, S)$, exactly what the note leg was sold as. An option leg's 0.107143 is worth 450, which is $\max(0, x_T - S)$. Nothing is left over and nothing is missing: the eighty notes take 71.428571 asset units and the eighty option legs take 8.571429. In the escrow's own fixed point each share is floored separately, so a redemption leg by leg leaves the rounding of chapter 4.2 behind, and it stays with the escrow rather than with the last holder out.

$$71.428571 + 8.571429 = 80$$

$80 * 3750 + 80 * 450 = 336000 = 80 * 4200$ the escrow ends empty before rounding
 notes take 71.428571 asset units, option legs take 8.571429

The put series mirrors it exactly. The nearest strike below the mark is 2,400, a deposit is 2,400 in cash, and the same 100 minted and 20 merged leave 80 pairs against 192,000 in the escrow. With the median settling at 2,000, each note leg takes 0.833333 of its deposit, which is 2,000 in cash, and each option leg takes 0.166667, which is 400. Those are $\min(x_T, S)$ and $\max(0, S - x_T)$ again, on the other side of the kernel.

$$160000 + 32000 = 192000$$

$80 * 2000 + 80 * 400 = 192000 = 80 * 2400$ the mirror, and the same shape

Neither series needed anyone to act at expiry, and neither divided a pot that depended on what holders chose to do. The only number either of them read is the one the window produced, and if the window had not produced one, the eighty pairs would still have been eighty deposits.